

Software Engineering II — Workshop No. 2

Context

TutorLink is a responsive web application designed to connect students seeking academic support with qualified independent tutors. The platform enables students to search for tutors by subject, availability, rating, and other criteria; book, reschedule or cancel sessions; communicate via in-app messaging; and provide ratings/reviews. Tutors manage their profiles, availability, and bookings. Administrators oversee accounts, disputes, and platform analytics.

This Workshop 2 document presents the **design phase artifacts** that bridge the requirements (Workshop 1) to implementation. The design directly supports all Functional Requirements (FR-01 to FR-12), Non-Functional Requirements (NFR-01 to NFR-05), and User Stories (US-S*, US-T*, US-A*) defined previously, with emphasis on the MVP Release 1 from the User Story Mapping (Onboarding → Profile Setup → Discovery → Scheduling → basic Session Delivery).

Key design decisions prioritize **clarity, maintainability, and alignment with NFRs** (performance target <2s for key actions, TLS encryption, WCAG 2.1 AA accessibility, 99.5% availability). The architecture uses a simple layered design (React + Python FastAPI backend) for simplicity in a student project while remaining easy to extend. The domain model uses inheritance for User roles and strong associations for core entities (Booking, Availability, Review).

CRC Cards

CRC Card: User (Base Class)

Responsibilities	Collaborators
• Handle user registration, authentication (login/logout) and credential management (email, password hash, role) • Provide common profile attributes (fullName, createdAt) and basic operations (updateProfile) • Support role-based access control (STUDENT, TUTOR, ADMIN) via getRole() • Serve as foundation for inheritance by Student and Tutor; centralizes security logic	Student, Tutor, Booking, Message, Review, Notification, StudentProfile, TutorProfile

CRC Card: Student

Responsibilities	Collaborators
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• Search and filter tutors (by subject, availability, rating, language) - supports US-S2, FR-04 • Book, reschedule or cancel tutoring sessions according to policy (US-S3, US-S4, FR-05, FR-06) • View personal session history and progress tracking (FR-12) • Submit ratings and reviews after completed sessions (US-S6, FR-10) • Manage learning preferences, goals and profile (FR-03)	User (inherits), Booking, Review, Message, Tutor (via search), Availability (read)
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CRC Card: Tutor

Responsibilities	Collaborators
• Create and maintain professional profile (bio, qualifications, subjects taught, hourlyRate) - FR-02, US-T1 • Define and update weekly recurring availability / time slots (US-T1) • Accept or decline incoming booking requests; control own schedule (US-T2) • View upcoming schedule and past session history (US-T4) • Receive notifications and messages; maintain averageRating from reviews	User (inherits), Booking, Availability, Review, Message, Subject (via tutor_subjects)

CRC Card: Booking

Responsibilities	Collaborators
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• Represent a concrete tutoring session instance (participants, time window, subject, status) • Manage lifecycle states: PENDING → CONFIRMED → COMPLETED / CANCELLED / RESCHEDULED (FR-05, FR-06) • Trigger notifications and reminders to both parties at key moments • Link to Review (post-completion) and contextual Messages; enforce cancellation policy	Student, Tutor, Availability (conflict check), Review, Message, Notification
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CRC Card: Availability

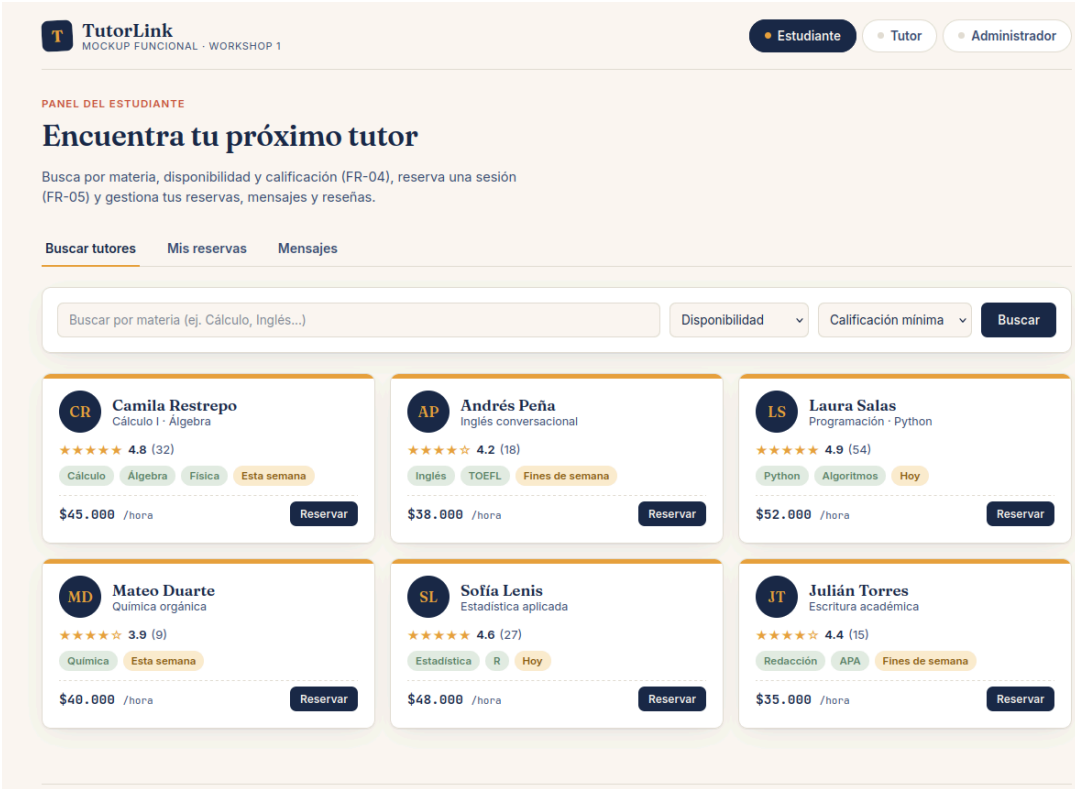
Responsibilities	Collaborators
• Define a tutor's available time windows (recurring by day-of-week or specific dates) • Support overlap detection to prevent double-booking when creating Booking • Provide queryable slots for student search and real-time calendar display in UI	Tutor, Booking (for conflict validation)

CRC Card: Review

Responsibilities	Collaborators
• Capture post-session student feedback: numeric rating (1-5) + optional comment (FR-10, US-S6) • Automatically update the associated Tutor's averageRating aggregate • Support moderation workflow (flag inappropriate reviews for Admin review - US-A4)	Student (author), Tutor (recipient), Booking (one-to-one link)

Mockups and Wireframes

They were created following usability principles (NFR-03: intuitive for non-technical users, WCAG 2.1 AA) and responsive design (NFR-05). Tools conceptually used: Figma/Balsamiq style; actual implementation would use React with accessible components. Each screen directly supports one or more user stories / functional requirements.



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TutorLink

MOCKUP FUNCIONAL · WORKSHOP 1

EstudianteTutorAdministrador

PANEL DEL ESTUDIANTE

Encuentra tu próximo tutor

Busca por materia, disponibilidad y calificación (FR-04), reserva una sesión (FR-05) y gestiona tus reservas, mensajes y reseñas.

Buscar tutoresMis reservasMensajes

TUTOR	MATERIA	FECHA	ESTADO	ACCIÓN
Camila Restrepo	Cálculo I	22 jul, 2026	Confirmada	Cancelar
Laura Salas	Python	25 jul, 2026	Pendiente	Cancelar
Andrés Peña	Inglés	15 jul, 2026	Completada	—
Mateo Duarte	Química	18 jul, 2026	Cancelada	—

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TutorLink

MOCKUP FUNCIONAL · WORKSHOP 1

EstudianteTutorAdministrador

PANEL DEL ESTUDIANTE

Encuentra tu próximo tutor

Busca por materia, disponibilidad y calificación (FR-04), reserva una sesión (FR-05) y gestiona tus reservas, mensajes y reseñas.

Buscar tutoresMis reservasMensajes

Hola, quería confirmar el tema de la próxima sesión: derivadas e integrales básicas. ¿Te parece bien?

Si, perfecto. ¿Podemos empezar repasando la regla de la cadena?

Claro, preparo unos ejercicios. Nos vemos el jueves.

Escribe un mensaje a Camila Restrepo...Enviar

Sesión completada

Cálculo I con Camila Restrepo — 12 jul, 2026

TU CALIFICACIÓN

★★★★★

Escribe una reseña breve...

Publicar reseña

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TutorLink

MOCKUP FUNCIONAL · WORKSHOP 1

EstudianteTutorAdministrador

PANEL DEL TUTOR

Gestiona tu perfil y tus clases

Edita tu perfil y disponibilidad (FR-02), acepta o rechaza solicitudes (US-T2) y revisa tu historial de sesiones (US-T4).

SolicitudesMi perfilHistorial

Solicitudes pendientes 2

ESTUDIANTE	MATERIA	HORARIO PROPUESTO	ESTADO	ACCIÓN
Julián Torres	Cálculo I	Jue 23 jul · 4:00pm	Pendiente	AceptarRechazar
Valeria Gómez	Álgebra Lineal	Vie 24 jul · 10:00am	Pendiente	AceptarRechazar

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TutorLink

MOCKUP FUNCIONAL · WORKSHOP 1

Estudiante

Tutor

Administrador

PANEL DEL TUTOR

Gestiona tu perfil y tus clases

Edita tu perfil y disponibilidad (FR-02), acepta o rechaza solicitudes (US-T2) y revisa tu historial de sesiones (US-T4).

Solicitudes

Mi perfil

Historial

Nombre completo

Camila Restrepo

Correo

camila.restrepo@tutorlink.co

Materias que enseño

Cálculo I, Álgebra Lineal, Física

Idioma

Español

Tarifa por hora (COP)

45.000

Calificación promedio

4.8 ★ (32 reseñas)

Cualificaciones

Ingeniera Matemática, Universidad Nacional. 5 años dando tutorías particulares en cálculo y física.

DISPONIBILIDAD SEMANAL

Lun	Mar	Mié	Jue	Vie	Sáb	Dom
9-11	9-11	9-11	9-11	9-11	9-11	9-11
2-4	2-4	2-4	2-4	2-4	2-4	2-4
5-7	5-7	5-7	5-7	5-7	5-7	5-7

Guardar cambios

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TutorLink

MOCKUP FUNCIONAL · WORKSHOP 1

Estudiante

Tutor

Administrador

PANEL DEL TUTOR

Gestiona tu perfil y tus clases

Edita tu perfil y disponibilidad (FR-02), acepta o rechaza solicitudes (US-T2) y revisa tu historial de sesiones (US-T4).

Solicitudes

Mi perfil

Historial

ESTUDIANTE	MATERIA	FECHA	ESTADO
Julián Torres	Cálculo I	12 jul, 2026	Completada
Valeria Gómez	Álgebra Lineal	08 jul, 2026	Completada
Mateo Duarte	Física	02 jul, 2026	Cancelada

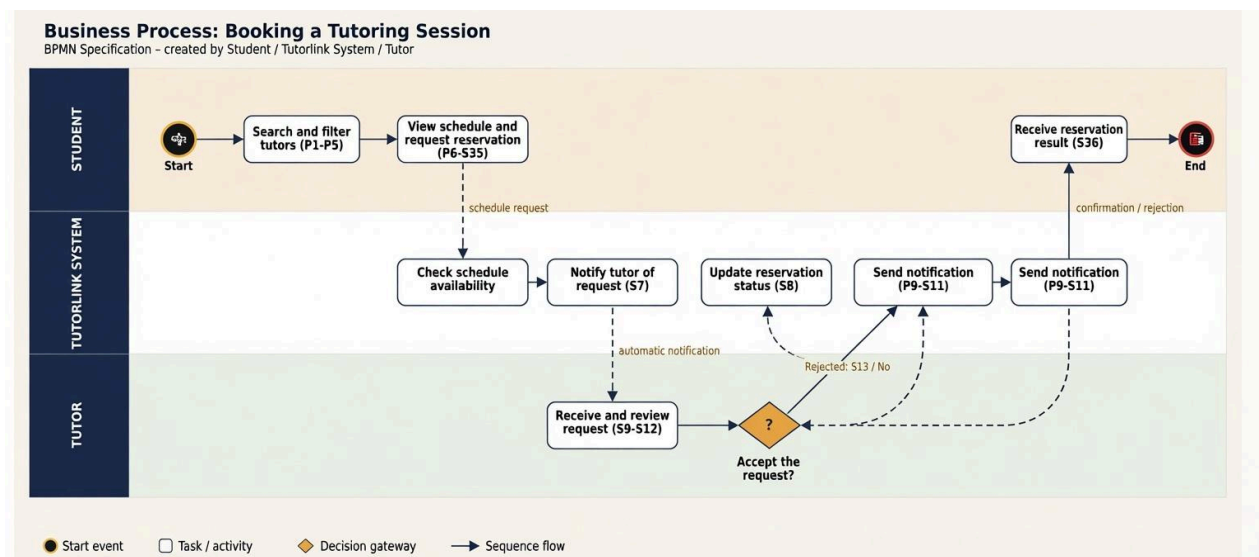


Business Model Processes

Core Process: Booking a Tutoring Session

The process begins when the student searches for and filters tutors by subject and availability (FR-04). Upon selecting a tutor, they view their available times and submit a booking request (US-S3). The system verifies that the time slot is still available and notifies the tutor of the new request (FR-07). The tutor reviews the request and decides whether to accept or reject it (US-T2) this is the only decision point in the process. With this response, the system updates the booking status (confirmed or rejected) and sends the corresponding notification to both parties. Finally, the student receives the result, completing the cycle.

Role in the application: This process is the functional backbone of TutorLink without it, there is no business, as it is the transaction that generates value for both the student (access to tutoring) and the tutor (revenue). Furthermore, it directly exposes the critical non-functional requirements: NFR-01 (availability checks and confirmations must be fast) and NFR-02 (booking data is transmitted encrypted). The other system flows (messaging, ratings, history) are extensions that depend on this process running correctly first.

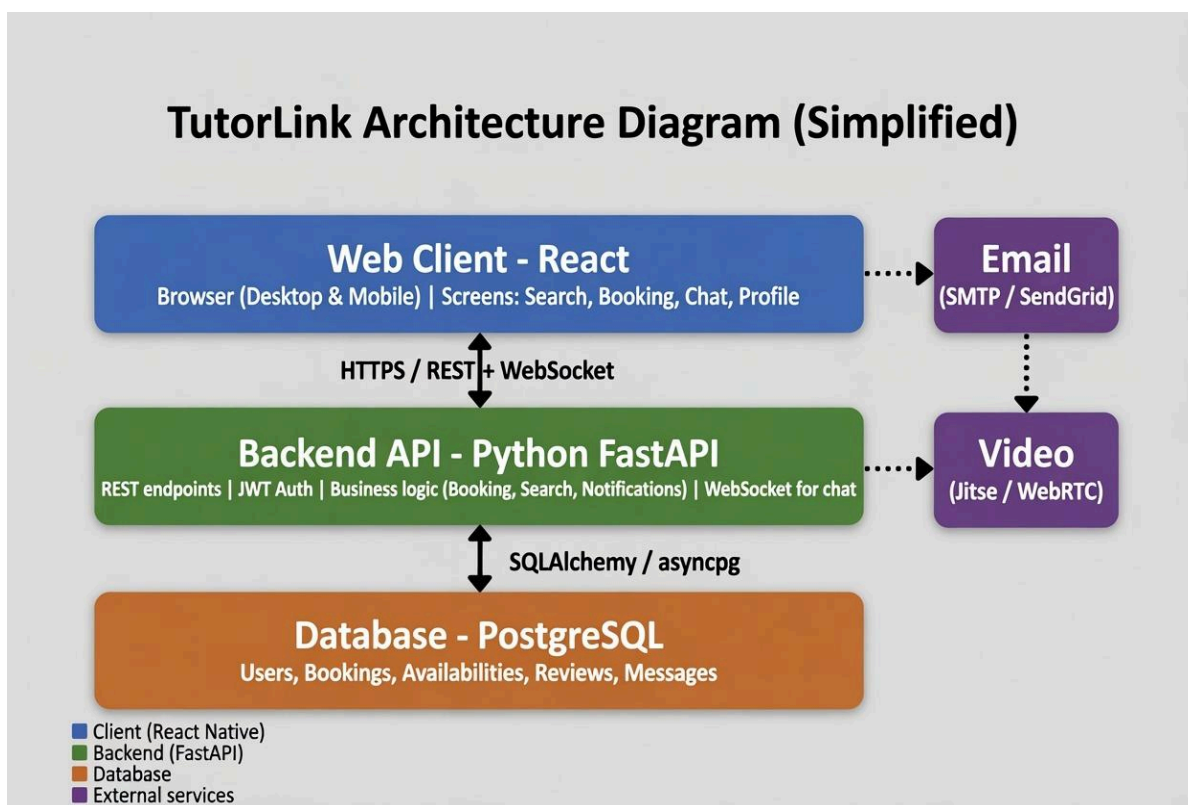


Architecture Diagram

The architecture is intentionally simple and layered to keep the project manageable for a student team while still supporting all functional and non-functional requirements. It consists of three clear layers plus a few external services.

Technology Stack (Simplified):

- **Client:** React — single codebase Handles all UI screens (search, booking calendar, chat, profiles, sessions history).
- **Backend API:** Python FastAPI — modern, fast, automatic OpenAPI docs, excellent async support and easy JWT authentication. Contains the business logic (booking state machine, search, notifications).
- **Database:** PostgreSQL — reliable relational store for users, profiles, availabilities, bookings, reviews and messages. Accessed via SQLAlchemy (or asyncpg).
- **Real-time:** WebSocket (supported natively by FastAPI) for in-app chat and live notifications.
- **External services:** SMTP / SendGrid for emails, Jitsi Meet or WebRTC for video sessions.
- **Auth:** JWT tokens (stateless), password hashing with bcrypt or Argon2.

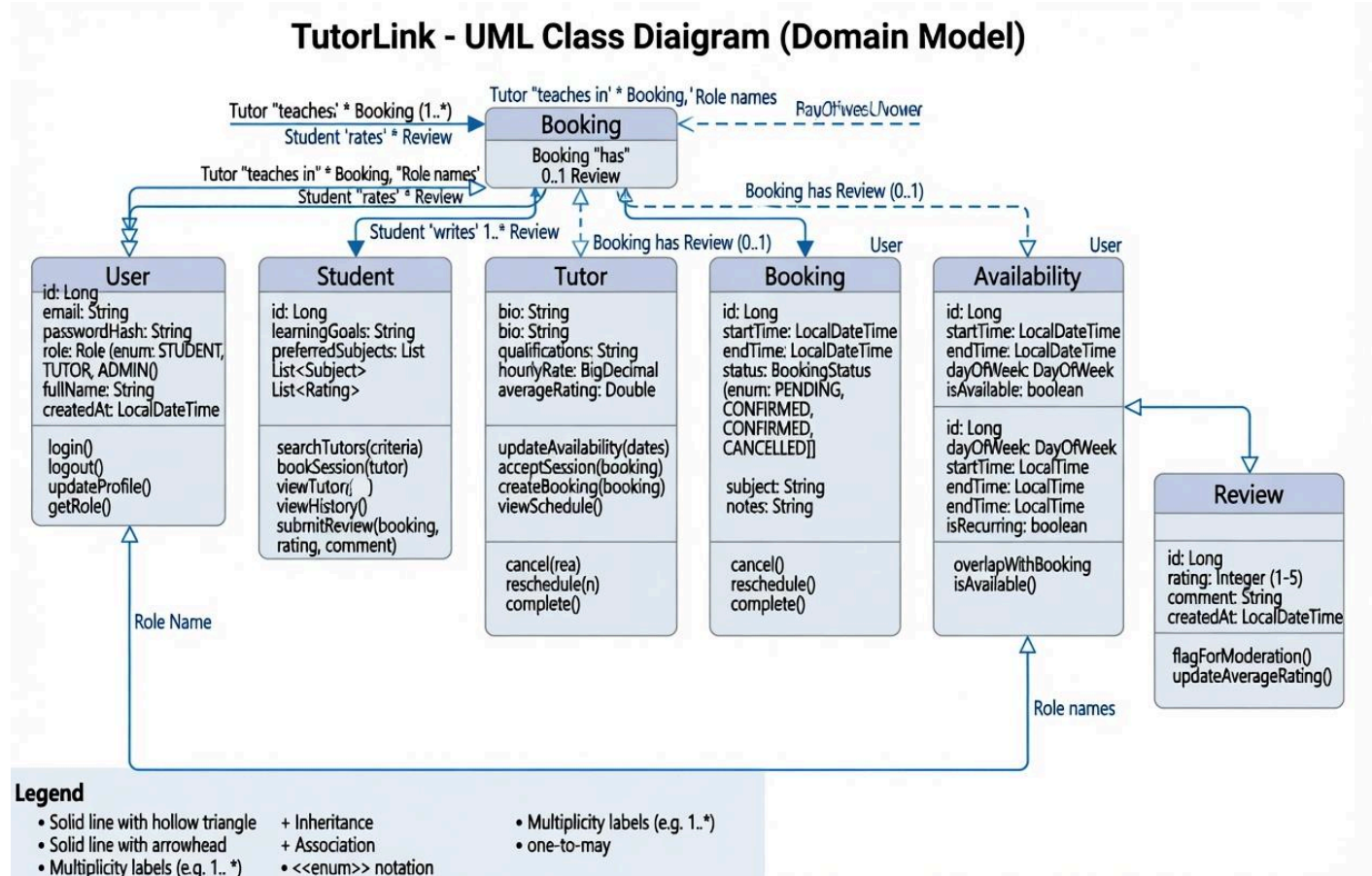


Communication is straightforward: React talks to the FastAPI backend over HTTPS (REST) for normal operations and over WebSocket for chat/notifications. The backend talks to PostgreSQL for persistence and calls external services only when needed (emails, video room creation). This design is easy to understand, implement and later extend.

Class Diagram

The UML Class Diagram below shows the core domain model classes, their attributes, key methods, inheritance (Student/Tutor extend User), and primary associations with multiplicities. It realizes the responsibilities identified in the CRC cards and supports the Booking process, search, reviews, and profile management from the requirements.

Note on associations: Student books many Bookings; Tutor participates in many Bookings and owns many Availabilities; Booking optionally has one Review; Review updates Tutor aggregate rating (via service). Subject is handled via many-to-many with junction (not shown for diagram clarity but present in DB). Methods are representative; full implementation would include getters/setters and additional validation logic.



Relational Database Model

Entity-Relationship Diagram

The ER diagram presents the normalized relational schema (targeting 3NF). It supports efficient queries for tutor search (via subjects + availabilities + ratings), booking conflict checks, and review aggregation. JSONB columns are used sparingly for flexible fields (e.g. preferred_schedule) while keeping core data relational for integrity and indexing.

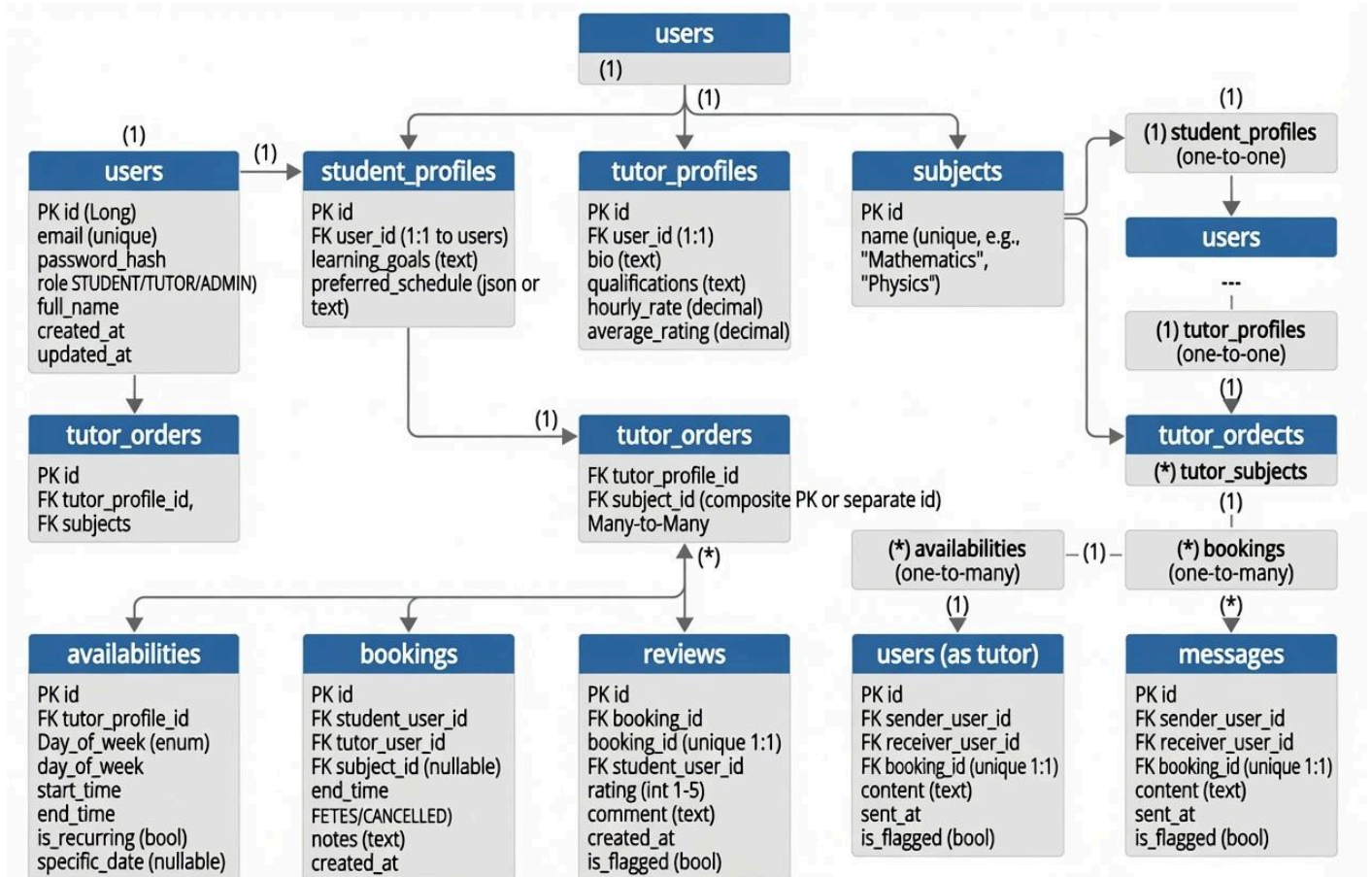


Figure 7: TutorLink Relational Database ER Diagram — Entities, primary/foreign keys, crow's foot relationships, and junction table for many-to-many (tutor_subjects).

7.2 Table Definitions and Key Constraints

Below are the main table definitions (PostgreSQL syntax). Full DDL would include indexes on frequently filtered columns (email, booking start_time, tutor search predicates), check constraints (rating 1-5, end_time > start_time), and triggers for average_rating updates.

users

id (PK), email (UNIQUE NOT NULL), password_hash, role (ENUM), full_name, created_at, updated_at

student_profiles

id (PK), user_id (FK → users, UNIQUE 1:1), learning_goals (TEXT), preferred_schedule (JSONB)

tutor_profiles

id (PK), user_id (FK → users, UNIQUE 1:1), bio (TEXT), qualifications (TEXT), hourly_rate (NUMERIC), average_rating (NUMERIC DEFAULT 0)

subjects

id (PK), name (UNIQUE NOT NULL) -- e.g. 'Mathematics', 'English'

tutor_subjects

tutor_profile_id (FK), subject_id (FK) -- composite or surrogate PK; many-to-many

availabilities

id (PK), tutor_profile_id (FK), day_of_week (ENUM), start_time, end_time, is_recurring (BOOL), specific_date (DATE NULLABLE)

bookings

id (PK), student_user_id (FK), tutor_user_id (FK), subject_id (FK NULLABLE), start_time, end_time, status (ENUM), notes (TEXT), created_at

reviews

id (PK), booking_id (FK UNIQUE 1:1), student_user_id (FK), rating (INT CHECK 1-5), comment (TEXT), created_at, is_flagged (BOOL
DEFAULT false)

messages

id (PK), sender_user_id (FK), receiver_user_id (FK), booking_id (FK NULLABLE), content (TEXT), sent_at, is_read (BOOL)

Key Relationships & Integrity: Foreign keys with ON DELETE RESTRICT or SET NULL as appropriate (e.g., keep historical bookings even if user deactivated). Booking status enum enforced at DB level. Triggers or application logic maintain tutor average_rating on review insert/update. Indexes recommended on (tutor_profile_id, day_of_week) for availability, and GIN on subjects for search if using array/JSON, but junction table is preferred for relational integrity.

8. References and Tools

[1] Workshop 2 instructions and templates provided by Eng. Carlos Andrés Sierra & Eng. Liliana Marcela Olarte (Universidad Nacional de Colombia, 2026).
[2] Workshop 1 deliverables: TutorLink Requirements, User Stories, and Story Mapping document.
[3] UML Diagrams overview: <https://www.uml-diagrams.org/class-diagrams-overview.html> (referenced for class diagram notation). [4] BPMN Guide: <https://www.visual-paradigm.com/guide/bpmn/what-is-bpmn/> (for business process modeling). [5] Lucidchart Architectural Diagrams guidance: <https://www.lucidchart.com/blog/how-to-draw-architectural-diagrams>. [6] FastAPI + React Native + PostgreSQL architecture patterns (standard industry references for similar mobile tutoring/booking platforms). [7] WCAG 2.1 AA accessibility guidelines and responsive design best practices (NFR-03, NFR-05 compliance). [8] Colombian data protection considerations (Ley 1581 de 2012 - Habeas Data) reflected in NFR-02 security design.

Diagrams in this document were created with AI-assisted generation (Grok Imagine) guided by precise prompts for UML/ER/wireframe notation, followed by visual review and minor corrections.